NEURAX 2.0 – Abstract

1. **Team Name:** HackStorm
2. **Theme Name:** HealthTech & MedAI
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5. **Project Title:** AI-Based Rural Health Monitoring and Early Disease Risk Prediction System
6. **Theme Selected:** HealthTech & MedAI

7. **Problem Statement:**
In many rural and underserved regions, access to timely and quality healthcare services remains a significant challenge. Limited availability of medical professionals, long travel distances to hospitals, and inadequate medical infrastructure often delay diagnosis and treatment. As a result, patients may experience worsening health conditions that could have been prevented through early intervention. Hospitals also face difficulties in managing patient inflow and prioritizing critical cases effectively. Furthermore, with the increasing digitization of healthcare systems, maintaining the privacy and security of sensitive medical records has become a critical concern.

8. **Proposed Solution:**
This project proposes an AI-powered remote healthcare platform that assists in early disease detection and health risk monitoring. Patients can input symptoms, vital parameters, and basic health information through a simple digital interface. Machine learning models analyze this data to predict potential health risks and categorize patient urgency into Emergency, Urgent, or Routine levels. The platform also provides a hospital management dashboard that helps healthcare providers monitor patient requests and allocate medical resources efficiently.

9. **Technical Approach:**
The system will utilize machine learning algorithms such as Random Forest or Logistic Regression for health risk prediction and triage classification. A Python-based backend using Flask or FastAPI will manage data processing and model integration. The user interface will be developed using Streamlit or web technologies including HTML, CSS, and JavaScript. Patient data will be stored securely in databases such as MongoDB or Firebase, with encryption mechanisms to protect sensitive information. Explainable AI techniques will be incorporated to highlight the key factors influencing predictions.

10. **Expected Impact:**
The proposed platform aims to improve early healthcare intervention by providing accessible digital health screening tools for people in remote areas. By combining predictive analytics with hospital workflow support, the system can help healthcare providers identify high-risk patients quickly and allocate resources more effectively. The architecture is designed to be scalable, enabling future integration with telemedicine services and broader healthcare networks, ultimately improving healthcare accessibility and efficiency.